Artificial Intelligence

An Introduction to Artificial Intelligence, Machine Learning and Deep Learning

Ms. Rubab (IWRM)

U.S.-Pakistan Center for Advanced Studies in Water
Mehran University of Engineering and Technology Jamshoro, Sindh, Pakistan



Learning Objectives

- Machine Learning and its Types
- Deep Learning and its Types

Artificial Intelligence

Artificial Intelligence is the field of creating computer systems that can perform tasks requiring human-like abilities such as learning, reasoning, perception, and decision-making.

1. 🗺️ Google Maps

AI → Traffic + Route Prediction → Best Route

2. 📱 Face Unlock

📷 Camera → AI/Face Recognition → 🔓 Phone Unlocked





Machine Learning

Machine Learning is a type of AI that allows computers to learn from data and make predictions or decisions without being directly programmed for every task.

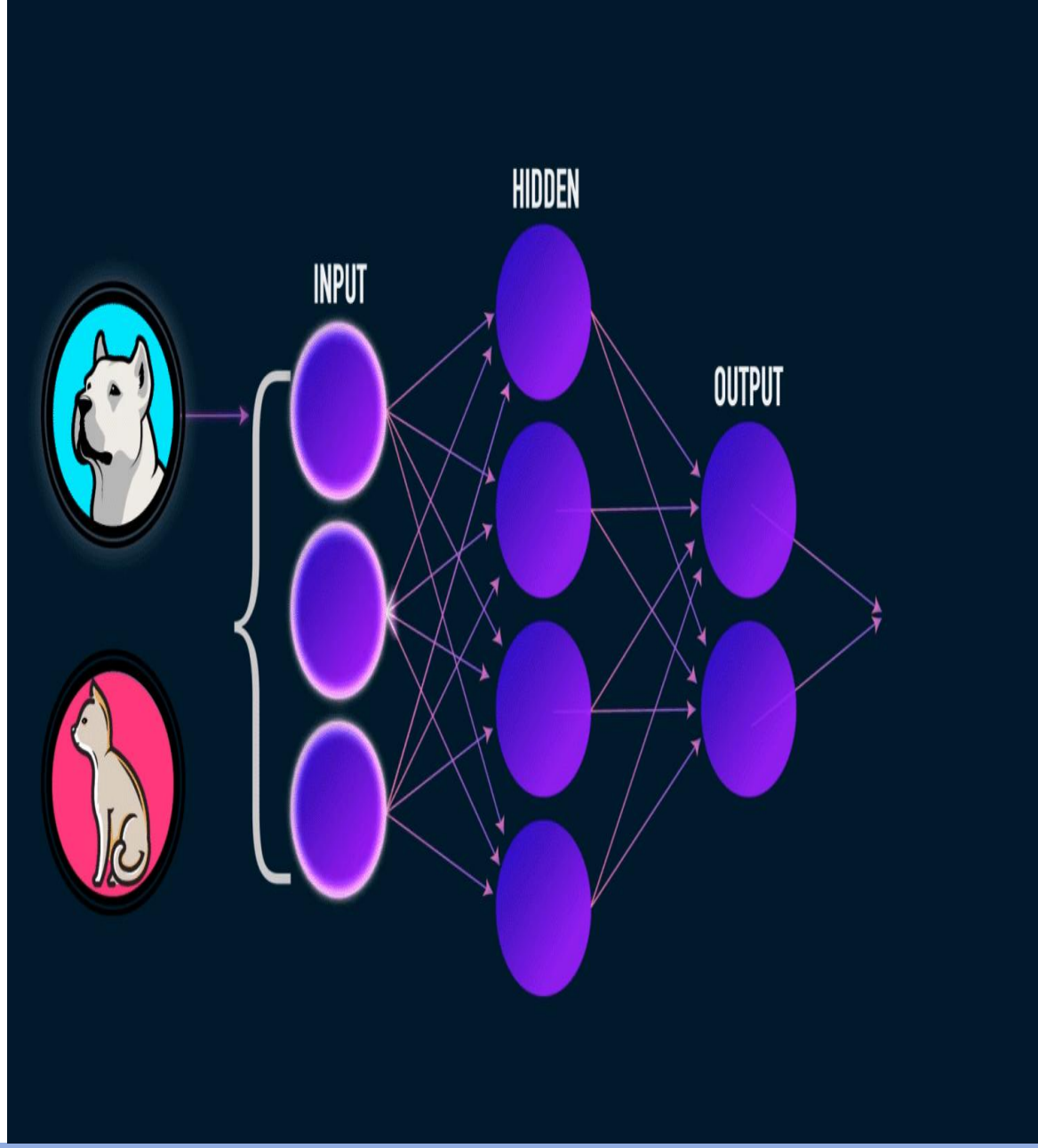
Recommending shows based on what you watch.

Online shopping sites showing you products you might like.

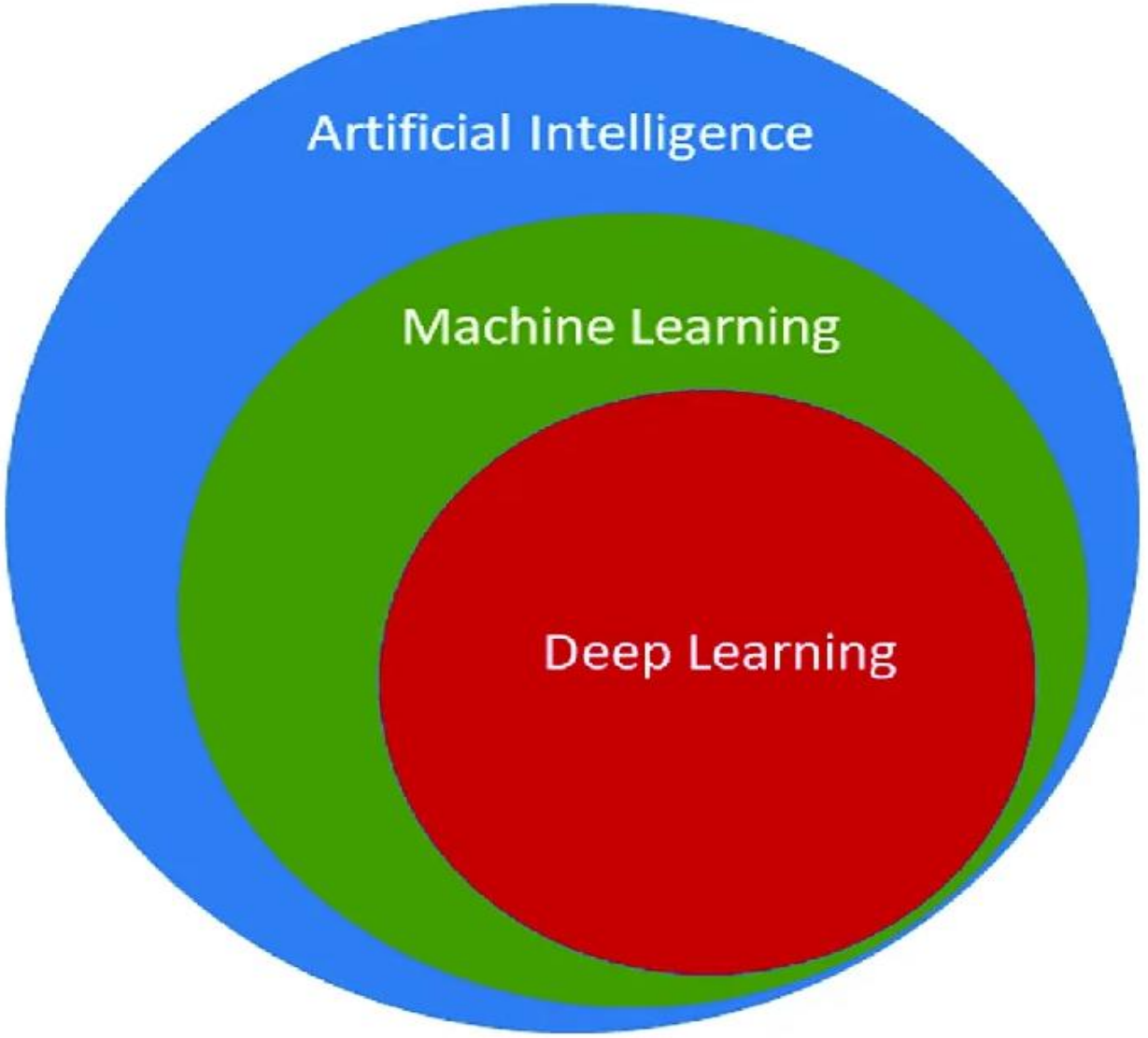
Deep Learning

Deep Learning is a type of Machine Learning that uses neural networks with multiple layers to learn complex patterns from large amounts of data.

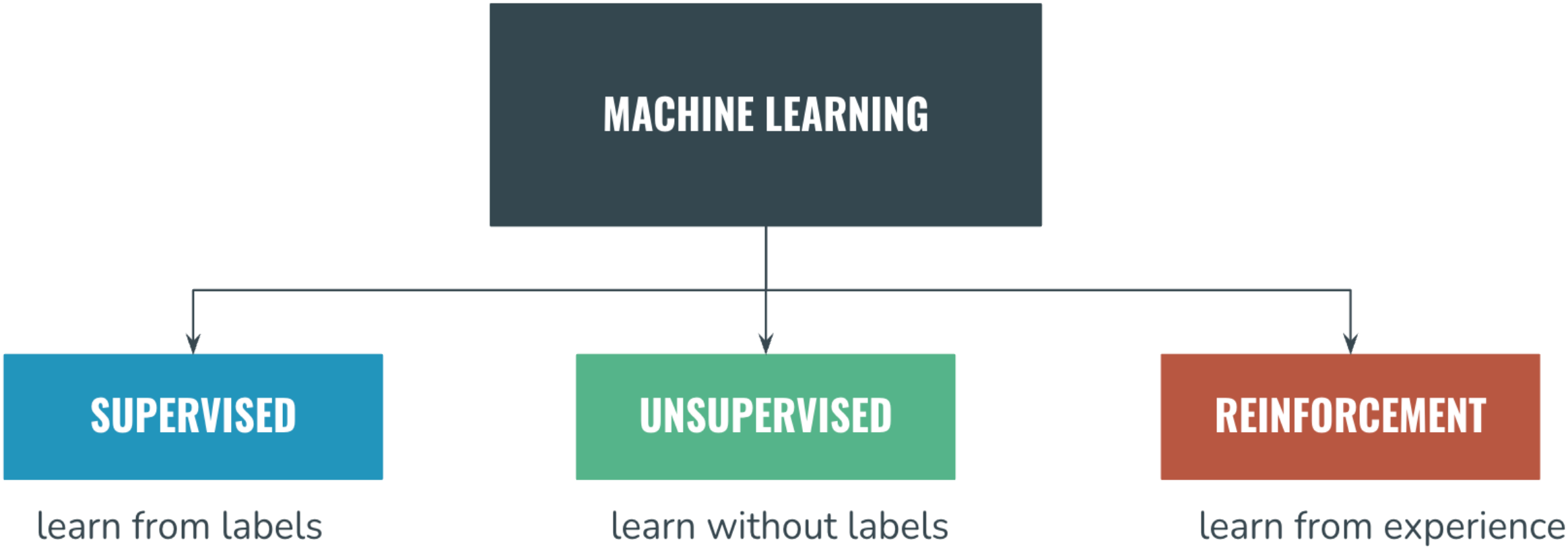
📷 Image → 🧠 Deep Learning → 👤 Face Recognition



- AI – Whole Brain
- ML – Teaching the brain to learn from experience
- DL – Teaching the brain by showing it lots and lots of examples



TYPES OF ML


MACHINE LEARNING
SUPERVISED
UNSUPERVISED
REINFORCEMENT
learn from labels
learn without labels
learn from experience

TYPES OF ML

1. 🎓 Supervised Learning

Model learns from labeled data.

Example: Predicting river stage from temperature, rainfall, soil moisture, and evaporation.

ML Models: **Regression, Classification**

Regression: Predicts a continuous numerical value. 📈

Example: Predicting river stage = 5.8 m.

Classification: Predicts a category or class. 🏷️

Example: Flood risk = Low, Medium, or High.

Supervised Learning: Common Algorithms

📈 **Linear Regression** → Predicts numerical values

🔢 **Logistic Regression** → Predicts categories

🌳 **Decision Tree** → Makes decisions using a tree structure

🎄 **Random Forest** → Uses many decision trees together

🎯 **KNN (K-Nearest Neighbors)** → Predicts using similar data points

📐 **SVM (Support Vector Machine)** → Separates data into classes

2. 🔍 Unsupervised Learning

Model finds hidden patterns or groups in data.

Example: Grouping areas based on similar rainfall or temperature patterns.

ML Models: **Clustering, Dimensionality Reduction**

Clustering: Groups similar data points together without predefined labels. 🔍

Example: Grouping areas with similar rainfall patterns.

Dimensionality Reduction: Reduces the number of variables while keeping the most important information. 📉

Example: Reducing 20 climate variables to 3 important ones.

Unsupervised Learning: Common Algorithms

🔵 **K-Means Clustering** → Groups similar data

🌳 **Hierarchical Clustering** → Creates groups in a hierarchy

🔎 **DBSCAN** → Finds natural groups and unusual points

📉 **PCA (Principal Component Analysis)** → Reduces many variables into fewer important variables

Example:

🌧️ Rainfall + 🌡️ Temperature + 💧 Soil Moisture → 🤖 ML → Similar Climate Groups

3. 🎮 Reinforcement Learning

Reinforcement learning is a branch of machine learning in which an AI agent learns to make optimal decisions through trial and error to maximize cumulative reward.

Example:

🚗 Correctly stops at a red light → ✅ Reward

🚗 Makes a wrong move → ❌ Penalty

Reinforcement Learning: Common Algorithms

🎯 **Q-Learning** → Learns the best action through rewards

🔄 **SARSA** → Learns from actions and their rewards

🧠 **DQN (Deep Q-Network)** → Uses deep learning for decision-making

Example:

🤖 Robot → 🚶 Action → ⭐ Reward / ❌ Penalty → 🧠 Learns the best action

TYPES OF DL

🧠 **ANN – Artificial Neural Network**

Learns patterns from data to make predictions.

Example: Predicting house prices from area, number of rooms, and location.

🖼️ **CNN – Convolutional Neural Network**

Learns from images and recognizes objects or patterns.

Example: Identifying whether an image contains a cat 🐱 or a dog 🐶.

🔄 RNN – Recurrent Neural Network

Learns from data that comes in sequence.

Example: Predicting tomorrow's temperature using previous days' temperatures.

⏳ LSTM – Long Short-Term Memory

Learns from sequences while remembering important past information.

Example: Predicting river stage using previous river stages, rainfall, temperature, and soil moisture.


Thank You